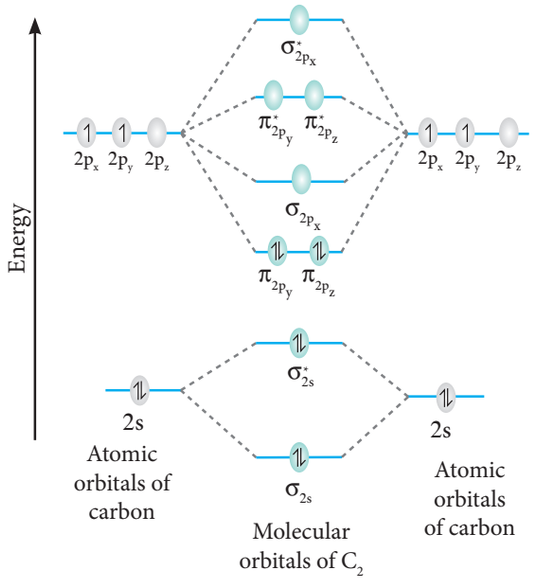


Fig 10.33 MO Diagram for C_2 molecule

Molecular orbital diagram of carbon molecule (C_2)

Electronic configuration of C atom is

$1s^2\,2s^2\,2p^2$

Electronic configuration of C_2 molecule

$\sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2, \sigma_{2s}^{*2}, \pi^2_{2p_y}, \pi^2_{2p_z}$

$$\text{Bond order} = \frac{N_b - N_a}{2} = \frac{8-4}{2} = 2$$

Molecule has *no unpaired electrons.* Hence, it is **diamagnetic.**

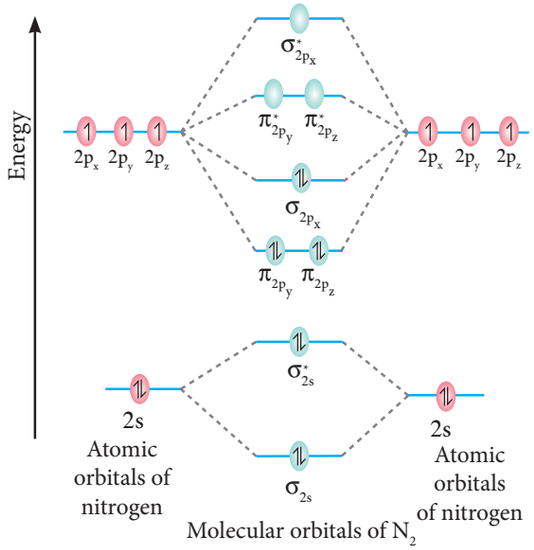


Fig 10.34 MO Diagram for N_2 molecule

Molecular orbital diagram of nitrogen molecule (N_2)

Electronic configuration of N atom is

$1s^2\,2s^2\,2p^3$

Electronic configuration of N_2 molecule

$\sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2, \sigma_{2s}^{*2}, \pi^2_{2p_y}, \pi^2_{2p_z} \sigma_2^2 px$

$$\text{Bond order} = \frac{N_b - N_a}{2} = \frac{10-4}{2} = 3$$

Molecule has *no unpaired electrons.* Hence, it is **diamagnetic.**

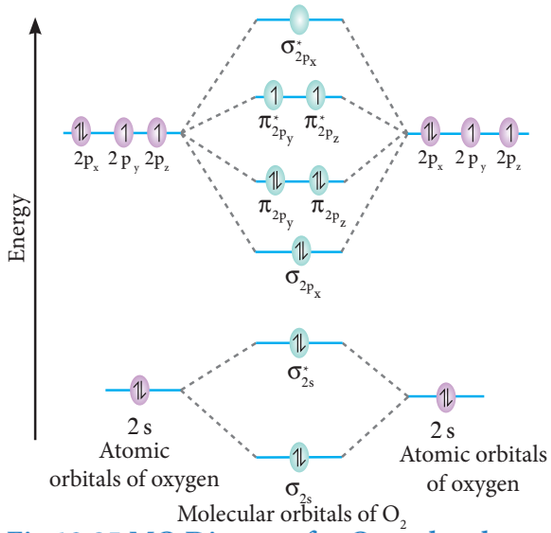


Fig 10.35 MO Diagram for O_2 molecule

Molecular orbital diagram of oxygen molecule (O_2)

Electronic configuration of O atom is

$1s^2\,2s^2\,2p^4$

Electronic configuration of O_2 molecule

$\sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2, \sigma_{2s}^{*2}, \sigma^2_{2p_x}\ \pi^2_{2p_y}, \pi^2_{2p_z}, \pi^{*1}_{2p_y}, \pi^{*1}_{2p_z},$

$$\text{Bond order} = \frac{N_b - N_a}{2} = \frac{10-6}{2} = 2$$

Molecule has *two unpaired electrons.* Hence, it is **paramagnetic**.

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